AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 26 · STUDENT LESSON

Embeddings and neighbourhoods

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Why should numerical differences between token IDs not be treated as differences in meaning?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Look up a vector using a token ID and preserve sequence shape. Try the recall before opening notes.

Your learning target

I can look up a vector using a token ID and preserve sequence shape.

I can calculate nearest neighbours with an explicit distance and tie rule.

I can distinguish constructed vectors from learned features and limit similarity claims.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Training** a small image model. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Image tensors and channels. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Look up a vector using a token ID and preserve sequence shape

Calculate nearest neighbours with an explicit distance and tie rule

Distinguish constructed vectors from learned features and limit similarity claims

Retrieve what you already know

> Why should numerical differences between token IDs not be treated as differences in meaning?

My starting explanation

This week’s end goal

By the end, I can explain embeddings and **neighbourhood**s using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Embeddings and neighbourhoods case. Separate observations, given facts and assumptions before choosing an explanation.

Continue the fictional robot vocabulary and add WAIT as ID 5. Use the table below, chosen by the lesson author for arithmetic practice. These coordinates are not measured robot properties and were not learned from data.

Each token ID selects one row; the resulting two-component vector is its embedding in this example.

Symbol	ID	Vector
PAD	0	[0, 0]
UNK	1	[-1, -1]
GO	2	[2, 0]
TURN	3	[1, 1]
STOP	4	[0, 2]
WAIT	5	[0, 1]

The table shape is 6 by 2: six vocabulary entries and **embedding dimension** 2. The ID sequence [2, 4, 0, 0] becomes [[2, 0], [0, 2], [0, 0], [0, 0]], with shape 4 by 2. Its mask remains [1, 1, 0, 0].

The vector dimension is not the sequence length or vocabulary size.

A zero PAD vector does not remove the need for masking. Averaging all four vectors above gives [0.5, 0.5]; averaging only the two real vectors gives [1, 1].

UNK is a real position when it replaces an unknown symbol, so its vector is not automatically excluded.

List the given facts and one fact that is not supplied.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Embeddings and neighbourhoods. Connect each labelled input, deciding step and result.

Use **squared Euclidean distance**: square each component difference and add. From GO [2, 0] to TURN [1, 1], the distance is (2 - 1) squared + (0 - 1) squared = 2. To STOP it is 8; to WAIT it is 5.

TURN is therefore the nearest of those three under this table and metric.

```
table = [[0, 0], [-1, -1], [2, 0],
         [1, 1], [0, 2], [0, 1]]
known = {"GO": 2, "TURN": 3, "STOP": 4, "WAIT": 5}

def distance2(a, b):
    if len(a) != len(b) or not a:
        raise ValueError("Need equal nonempty dimensions")
    return sum((x - y)**2 for x, y in zip(a, b))

def nearest(query, candidates):
    if not candidates:
        raise ValueError("Need candidates")
    scores = {name: distance2(query, vector)
              for name, vector in candidates.items()}
    best = min(scores.values())
    return sorted(name for name, d in scores.items() if d == best)

candidates = {name: table[i] for name, i in known.items()
              if name != "GO"}
assert nearest(table[2], candidates) == ["TURN"]
```

The candidate set excludes PAD, UNK and the query token itself. Otherwise a query would normally retrieve itself at distance zero.

This code assumes finite numeric components and uses exact ties, appropriate for these integer examples. Floating-point comparisons in other experiments need a declared precision policy.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Embeddings and neighbourhoods. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

For query [1, 0], compute squared distances to GO, TURN, STOP and WAIT in the original table. Return all nearest candidates.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

> **Change one thing**
>
> Change: Move the query to [2, 1].

Predicted result

Prediction: Q becomes nearest with squared distance1; P distance4; R distance10.

Why the result follows

Reason: Changing the query position changes geometric neighbours while the stored table remains fixed.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Embeddings and neighbourhoods to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Embedding table	A table assigning a fixed-length numerical vector to each token ID
Embedding dimension	The number of components in each embedding vector
Neighbourhood	Nearby candidates under a stated representation and distance rule
Squared Euclidean distance	The sum of squared component differences between two vectors

Use two terms to explain a step in this week's example.

Follow a worked example

Follow the worked trace for Embeddings and neighbourhoods, predict a step, then compare with its explanation.

One query can have two closest candidates

CANDIDATE VECTOR	QUERY VECTOR	SQUARED DISTANCE
GO [2, 0]	[1, 0]	1
TURN [1, 1]	[1, 0]	1
WAIT [0, 1]	[1, 0]	2
STOP [0, 2]	[1, 0]	5

The nearest set is GO and TURN. Alphabetical display order does not break the mathematical tie or make one candidate more similar.

Using STOP [0, 2] as the query, calculate distances to GO, TURN and WAIT. Exclude STOP itself and both reserved entries. Which candidate is nearest?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Embeddings and neighbourhoods

Measure embedding neighbours without turning numerical proximity into factual certainty.

The facts we will use

Supplied facts

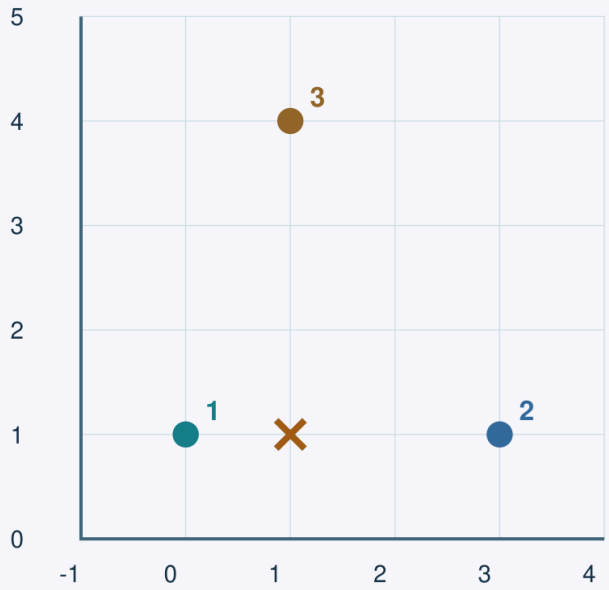
Toy tableP=[0, 1], Q=[3, 1], R=[1, 4]; query [1, 1]. Use **squared Euclidean distance**.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

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Squared Euclidean embedding distances



Horizontal: first embedding coordinate; vertical: second embedding coordinate

1: P at (0, 1)

2: Q at (3, 1)

3: R at (1, 4)

Cross: query at (1, 1)

Distance rule: Squared Euclidean distance

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/26

Embeddings and neighbourhoods

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Distance to P

(1-0)2+(1-1)2=1.



2. Distances to Q/R

Q distance4; R distance9.



3. Select neighbour

P is nearest under the declared metric; this is a geometric result, not proof of meaning.

CHANGE ONE THING

Move the query to [2, 1].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Distance to P	$(1-0)^2+(1-1)^2=1$.
2. Distances to Q/R	Q distance4; R distance9.
3. Select neighbour	P is nearest under the declared metric; this is a geometric result, not proof of meaning.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

An embedding stores each item as an ordered vector. Under **squared Euclidean distance**, subtract matching coordinates, square their differences and add. The smallest total identifies the nearest supplied vector.

These invented coordinates let us check geometry exactly; a claim that neighbours share meaning needs **evidence** about how the representation was learned and used.

What this example does not establish

Keep the stated limit

Embedding geometry depends on **training**, metric and scale; proximity does not prove truth,

causation or source permission.

Keep the stated limit

Read the labelled picture: What would justify claiming nearby vectors mean related topics? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Move the query to [2, 1]. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Embeddings and neighbourhoods claim and explain a repair using the supplied evidence.

Check the claim against the facts

An **embedding table** can be trainable: an optimisation procedure adjusts its entries to reduce a declared objective.

To isolate one update, let the GO row e start at [2, 0] and use the classroom loss J(e) = (e1 - 0) squared + (e2 - 2) squared.

The target vector [0, 2] is chosen for this arithmetic demonstration, not discovered **evidence** about the word GO.

Check the claim against the facts

The gradient at the start is [4, -4]. With learning rate 0.25, subtract the step to obtain e = [1, 1]. Loss decreases from 8 to 2. Keep every other table row frozen in this isolated example.

GO now has the same vector as TURN, so a downstream rule receiving only this vector cannot distinguish those two tokens.

Check the claim against the facts

This update proves only that the declared objective decreased. It does not prove that the new **neighbourhood** reflects useful meaning. Learned **neighbourhood**s depend on the **training** data, objective and model.

A nearby token need not be a safe substitute, have the same label, or be interchangeable in a sentence or command.

Locate the earliest unsupported step

For the isolated GO-row update, give the gradient, updated vector, starting loss and new loss. Which other token now shares the vector?

State one repair and the checks you would rerun.

Practise together

> Practise Embeddings and neighbourhoods with support, then explain the deciding step in your own words.

Using the original table, embed TURN WAIT STOP with one right-padding slot. Give the vector sequence, mask and shape.

> **Using STOP [0, 2] as the query, calculate distances to GO, TURN and WAIT. Exclude STOP itself and both reserved entries. Which candidate is nearest?**

> **Explain your prediction to a partner before comparing results.**

Practise with less help: Calculate distances and a tie

Use workbook W2 for the connected example “Embeddings and **neighbourhood**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Embeddings and **neighbourhood**s. Proposed explanation: “Token ID differences define embedding distances”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What would justify claiming nearby vectors mean related topics? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Embeddings and neighbourhoods. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

For query [1, 0], compute squared distances to GO, TURN, STOP and WAIT in the original table. Return all nearest candidates.

```
table = [[0, 0], [-1, -1], [2, 0],
         [1, 1], [0, 2], [0, 1]]
known = {"GO": 2, "TURN": 3, "STOP": 4, "WAIT": 5}

def distance2(a, b):
    if len(a) != len(b) or not a:
        raise ValueError("Need equal nonempty dimensions")
    return sum((x - y)**2 for x, y in zip(a, b))

def nearest(query, candidates):
    if not candidates:
        raise ValueError("Need candidates")
    scores = {name: distance2(query, vector)
              for name, vector in candidates.items()}
    best = min(scores.values())
    return sorted(name for name, d in scores.items() if d == best)

candidates = {name: table[i] for name, i in known.items()
              if name != "GO"}
assert nearest(table[2], candidates) == ["TURN"]
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Embeddings and neighbourhoods. Explain why each step is justified.

Explain → apply → challenge

Using the original table, embed TURN WAIT STOP with one right-padding slot. Give the vector sequence, mask and shape.

For the isolated GO-row update, give the gradient, updated vector, starting loss and new loss. Which other token now shares the vector?

A developer swaps the IDs of GO and STOP but leaves the embedding rows unchanged. What goes wrong, and what must be saved or changed together?

Explain which evidence supports your conclusion and which claim remains unproved.

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled embeddings and **neighbourhood**s **evidence** sheet to Image-pattern Network Investigation. Show how you can look up a vector using a token id and preserve sequence shape.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

In the original table, the masked average for GO STOP is [1, 1], and STOP GO has the same average. It also equals the single TURN vector.

Averaging discarded order and can collide across different sequence lengths unless additional information is retained. Keeping the sequence of vectors makes order available; a later model must still use it.

A teammate says TURN must mean the same as GO because it is the closest candidate, and says a lower embedding-training loss proves this. Explain what the calculations establish and what evidence is missing.

Save the vocabulary-to-row mapping, table version, mask convention, distance rule and candidate policy with the model. Compare embedding choices using **training** and validation discipline, and keep final evaluation reserved.

Next week examines how normalisation and similarity measures can change neighbour rankings.

A developer swaps the IDs of GO and STOP but leaves the embedding rows unchanged. What goes wrong, and what must be saved or changed together?

My limited claim and a useful next check

Study a complete case

> Revisit the complete Embeddings and neighbourhoods case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

One query can have two closest candidates

CANDIDATE VECTOR	QUERY VECTOR	SQUARED DISTANCE
GO [2, 0]	[1, 0]	1
TURN [1, 1]	[1, 0]	1
WAIT [0, 1]	[1, 0]	2
STOP [0, 2]	[1, 0]	5

The nearest set is GO and TURN. Alphabetical display order does not break the mathematical tie or make one candidate more similar.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Find a neighbourhood independently

A separate table contains A [0, 0], B [2, 0] and C [1, 2]. For query [1, 0], give all squared distances and every nearest candidate. Then calculate the mean of A and B and compare it with the query.

Can that mean identify which ordered pair produced it?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A separate table contains A [0, 0], B [2, 0] and C [1, 2]. For query [1, 0], give all squared distances and every nearest candidate. Then calculate the mean of A and B and compare it with the query.

Can that mean identify which ordered pair produced it?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Embeddings and neighbourhoods, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Measure embedding neighbours without turning numerical proximity into factual certainty.

By the end, I can	How I will show it
Look up a vector using a token ID and preserve sequence shape	Explain the deciding step, show a trace or test, and state one limit.
Calculate nearest neighbours with an explicit distance and tie rule	Explain the deciding step, show a trace or test, and state one limit.
Distinguish constructed vectors from learned features and limit similarity claims	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Embeddings and neighbourhoods without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Look up a vector using a token ID and preserve sequence shape

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Measure embedding neighbours without turning numerical proximity into factual certainty.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? P is nearest under the declared metric; this is a geometric result, not proof of meaning.

1. Distance to P

2. Select neighbour

3. Distances to Q/R

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Select neighbour P is nearest under the declared metric; this is a geometric result, not proof of meaning. An embedding stores each item as an ordered vector.

Under **squared Euclidean distance**, subtract matching coordinates, square their differences and add. The smallest total identifies the nearest supplied vector.

These invented coordinates let us check geometry exactly; a claim that neighbours share meaning needs **evidence** about how the representation was learned and used.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move the query to [2, 1]. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Q becomes nearest with squared distance1; P distance4; R distance10.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Q becomes nearest with squared distance1; P distance4; R distance10. Changing the query position changes geometric neighbours while the stored table remains fixed.

Repair a missing building block

Review Sequence representations (ai-5/25). Goal: Encode an ordered sequence with an explicit vocabulary. Short practice: Which record lets the program distinguish the first and last zeros?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** a small image model (ai-5/23). Goal: Trace a complete image model with fixed features and one trained weight. Short practice: Does lower **training** loss alone establish that four updates generalise better?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Image tensors and channels (ai-5/20). Goal: Read a pixel using declared row column and channel axes. Short practice: Where is the bottom green pixel in the first batched image?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Embeddings and neighbourhoods graphic using its labels, deciding evidence and relevant limitation.

01. Embeddings and neighbourhoods: the case



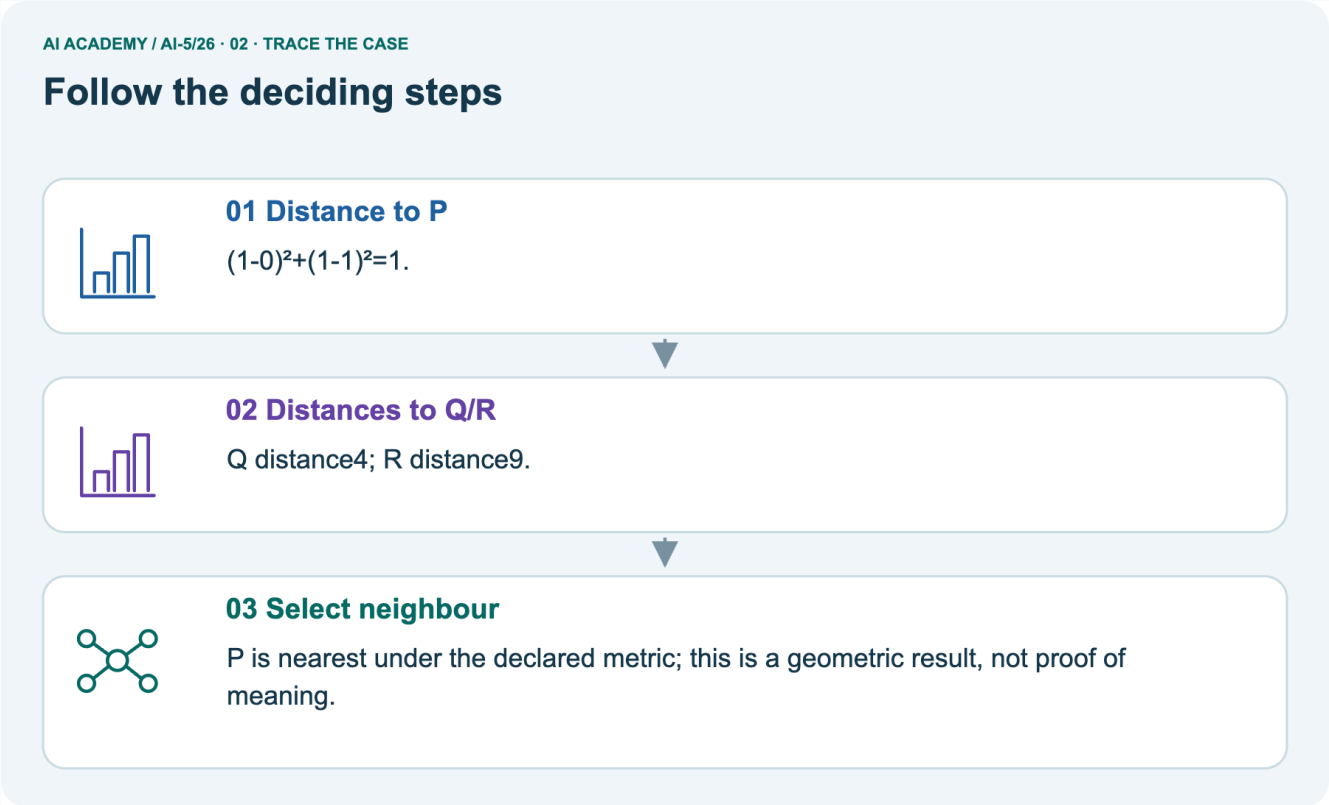
Follow the labelled picture with the matching explanation. Open the vector diagram

Measure embedding neighbours without turning numerical proximity into factual certainty. Supplied facts: Toy tableP=[0, 1], Q=[3, 1], R=[1, 4]; query [1, 1]. Use **squared Euclidean distance**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

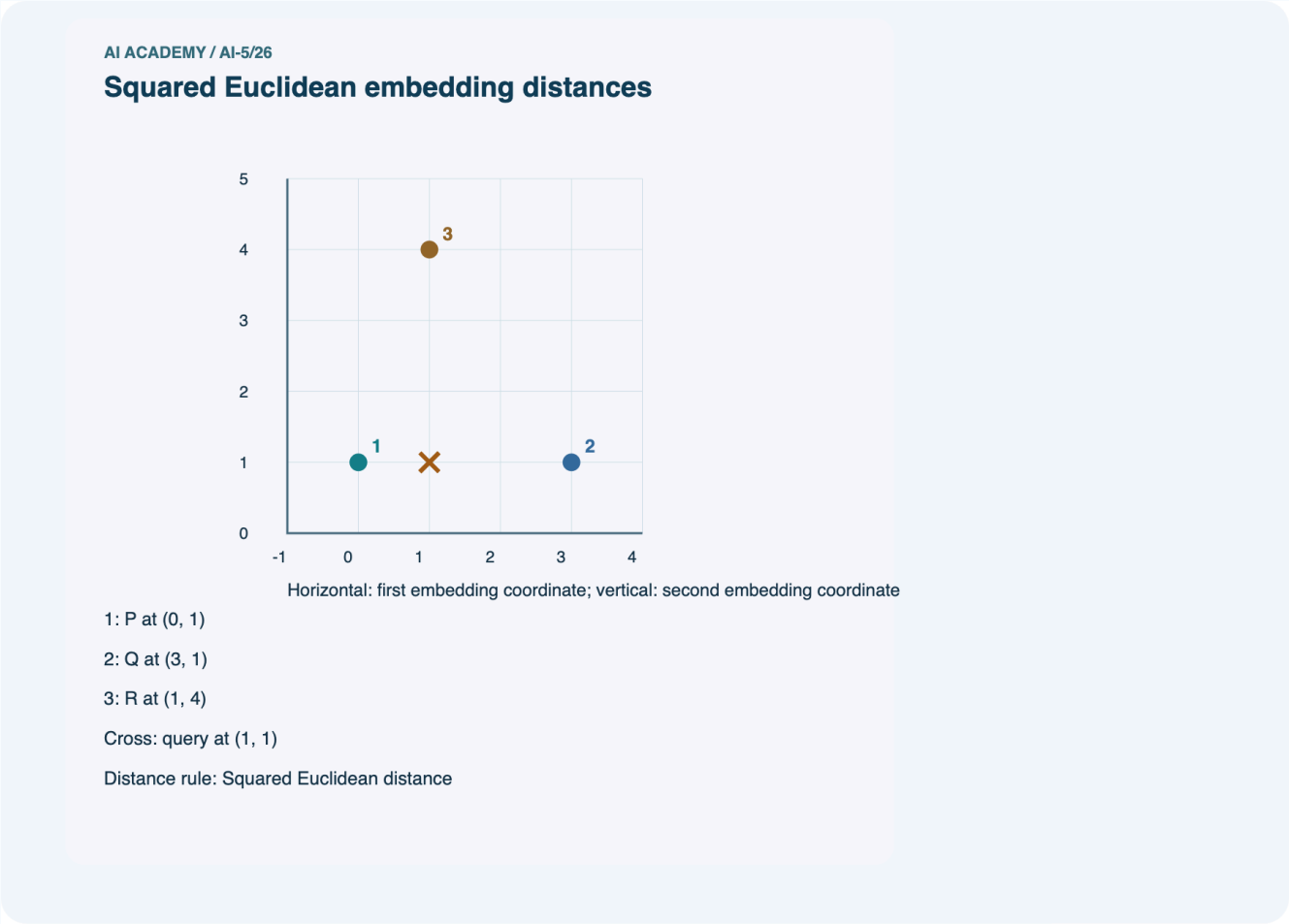


Follow the labelled picture with the matching explanation. Open the vector diagram

Distance to P: (1-0)²+(1-1)²=1. → Distances to Q/R: Q distance4; R distance9. → Select neighbour: P is nearest under the declared metric; this is a geometric result, not proof of meaning.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Squared Euclidean embedding distances

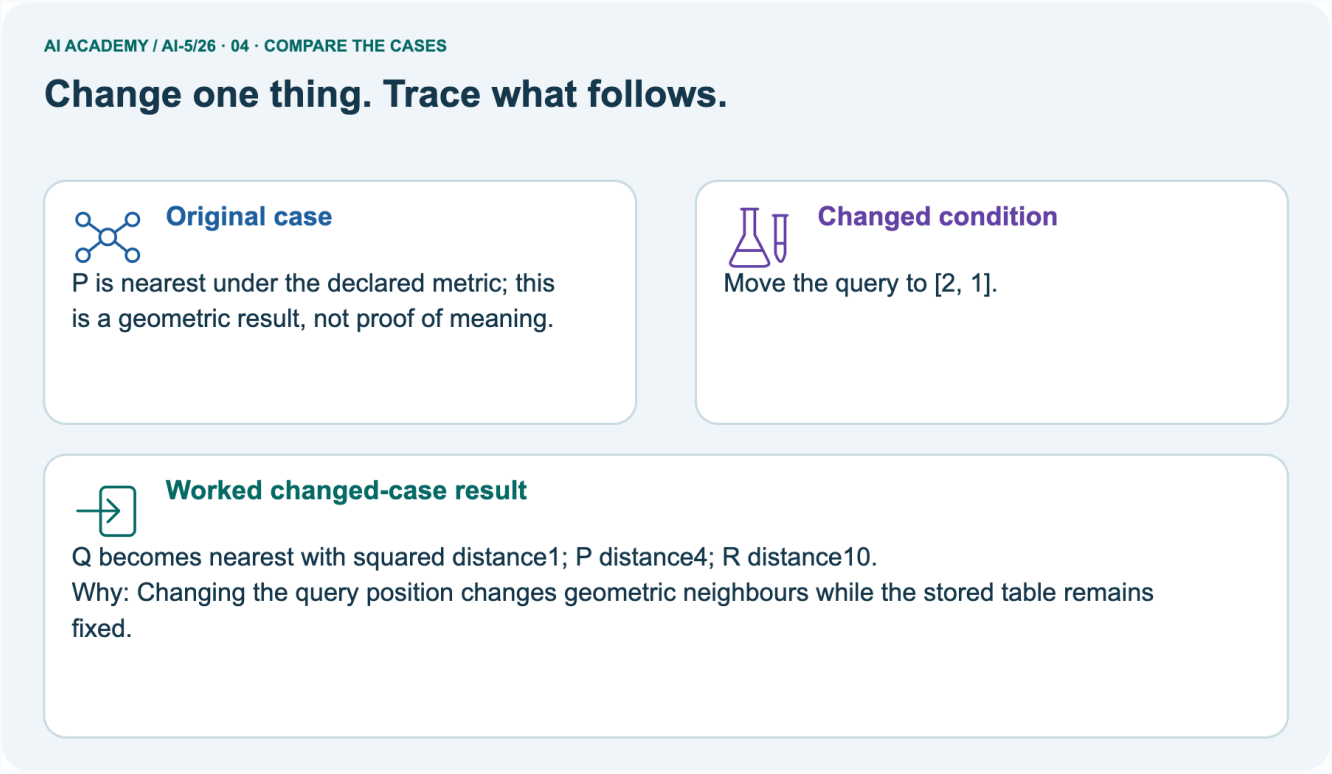


Follow the labelled picture with the matching explanation. Open the vector diagram

Squared Euclidean embedding distances. P: (0, 1). Q: (3, 1). R: (1, 4). Query query: (1, 1). Axes: first embedding coordinate; second embedding coordinate. Distance: **Squared Euclidean distance**.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



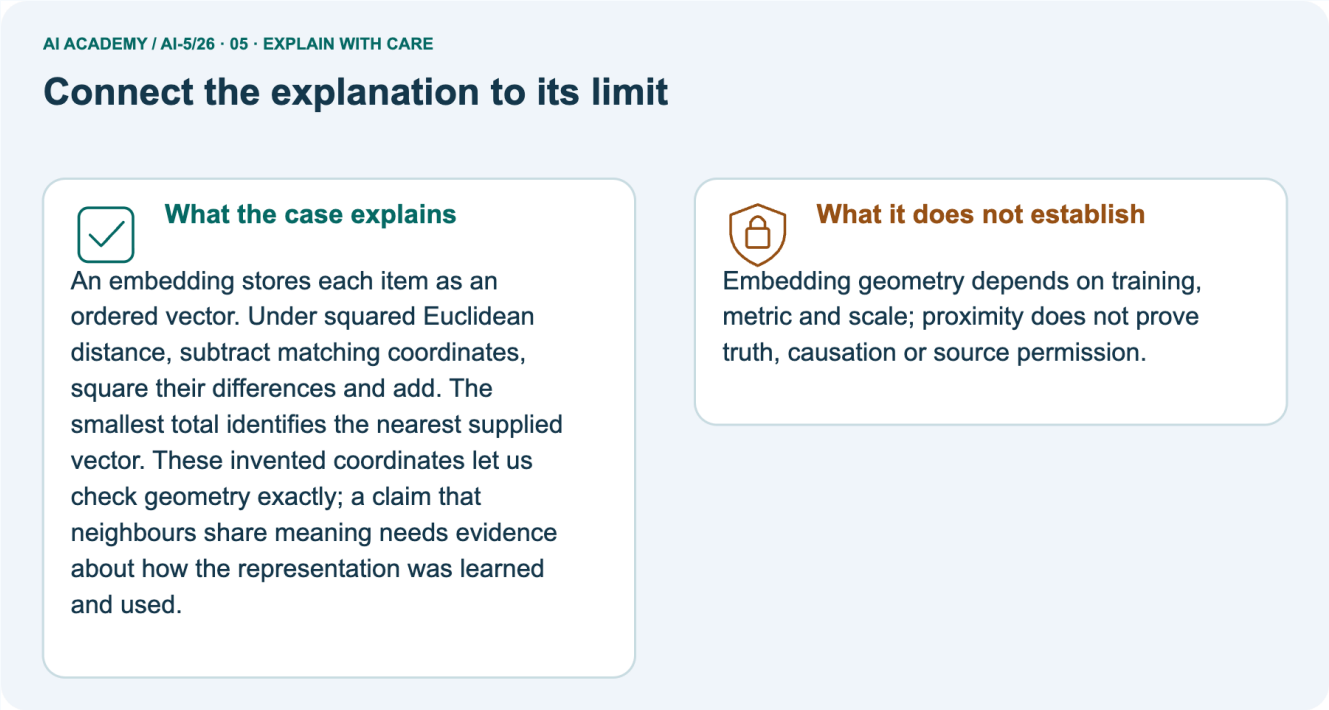
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: P is nearest under the declared metric; this is a geometric result, not proof of meaning. Changed condition: Move the query to [2, 1].

Expected result: Q becomes nearest with squared distance1; P distance4; R distance10. Reason: Changing the query position changes geometric neighbours while the stored table remains fixed.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

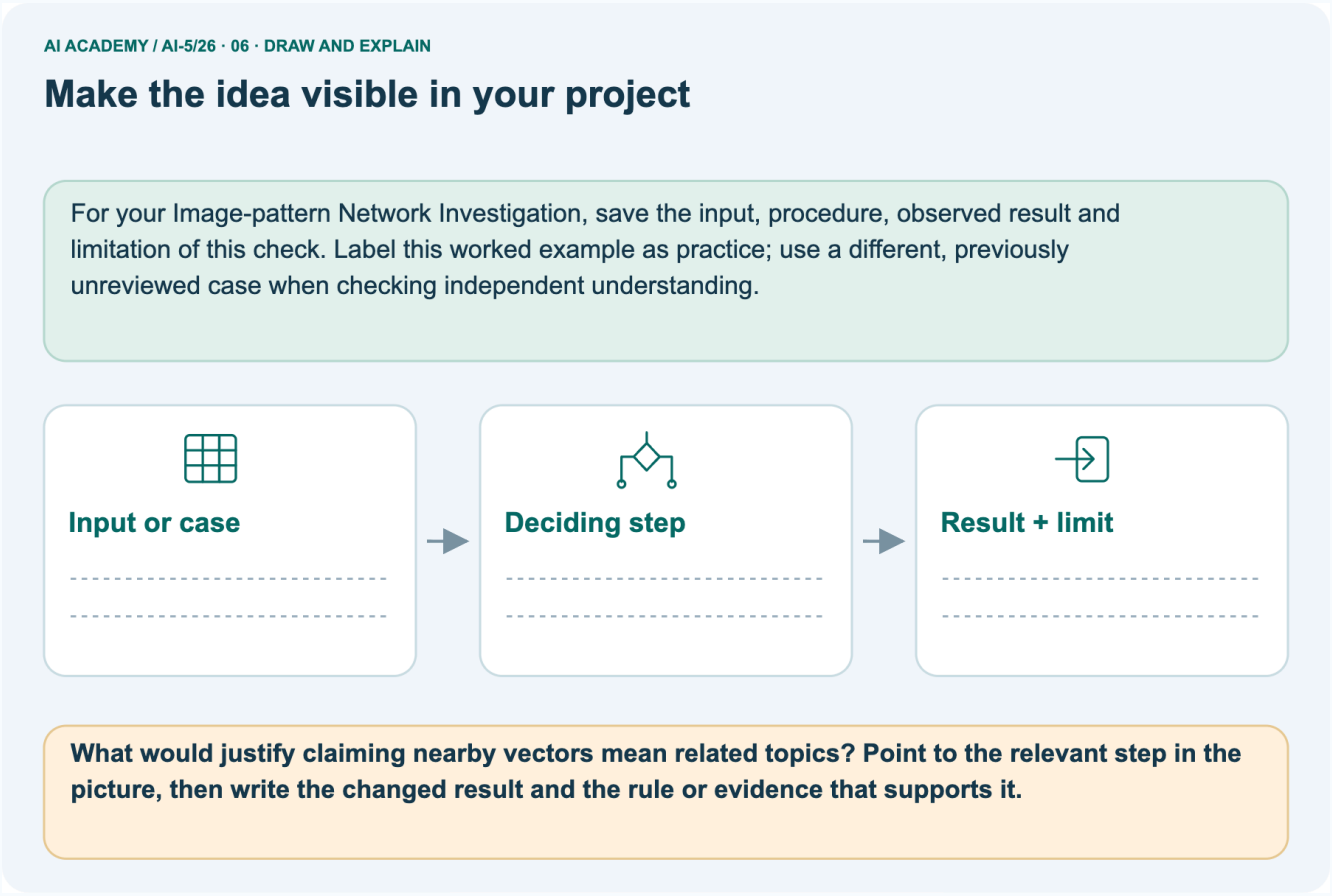
An embedding stores each item as an ordered vector. Under **squared Euclidean distance**, subtract matching coordinates, square their differences and add. The smallest total identifies the nearest supplied vector.

These invented coordinates let us check geometry exactly; a claim that neighbours share meaning needs **evidence** about how the representation was learned and used.

Limit: Embedding geometry depends on **training**, metric and scale; proximity does not prove truth, causation or source permission.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What would justify claiming nearby vectors mean related topics? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.